MA491 Project 3 Group Report

MA491 Project 3 Group 2

November 21, 2024

1 Introduction

We model the assignment of radio channels to transmitters on a hexagonal grid. We consider a variety of constraints on valid channel assignments to each transmitter based on proximity to other transmitters' values. We establish bounds on the span of frequencies required to create a valid assignment on both a finite and infinite grid subject to various constraints.

2 Terms

We define the following terms for our model.

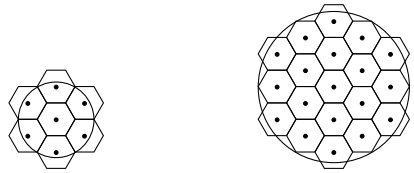
- Hexagon - one hexagonal cell in a grid, equivalent to a transmitter.
- Value - channel value assigned to a hexagon.
- Assignment - a function from a set of cells to a set of values.
- Valid assignment - an assignment that satisfies the constraints.
- Span - the highest value of an assignment covering an entire grid.

3 Constraints

We model radio channel assignments in a region partitioned into a grid of regular hexagons, with a transmitter at the center of each hexagon. We assign channels positive integers. Let s be the side length of a hexagon. We have two constraints on the assignment numbers to reduce interference between the transmitters:

1. Any two transmitters within distance $2s$ of each other must have channel numbers that differ by at least k_1.
2. Any two transmitters within distance $4s$ of each other must have channel numbers that differ by at least k_2.

Figure 1a shows the transmitters within distance $2s$ of a transmitter, and Figure 1b shows the transmitters within distance $4s$ of a transmitter.



(a) Constraint 1 (b) Constraint 2

Figure 1: Constraints

4 Observations

Since our goal is to minimize the span, we make some observations about minimum channel assignments.

Lemma 1

Any minimized channel assignment must contain some hexagon assigned the number 1.

Proof. Towards contradiction, assume there is some optimal channel assignment with minimum number $k \neq 1$. Then, a new assignment with each number decreased by $k - 1$ is equally valid, since the difference between any two hexagons is unchanged. The span of this new assignment would be smaller than the span of the original assignment, thus the new assignment was not minimized. Therefore any optimized channel assignment must contain some hexagon assigned the number 1. ⬢

We also notice that if one arrangement, A_1, of hexagons contains another arrangement, A_2, of hexagons, then the span of A_1 is at least the span of A_2.

5 Infinite Grid

An infinite grid of regular hexagons spreads arbitrarily far in all directions. We find upper and lower bounds for the case with $k_2 = 1$ if no two transmitters within distance $4s$ of each other can have the same channel.

5.1 $k_1 = 2$

For $k_1 = 2$, we determine upper and lower bounds on the span. For the upper bound, we find a repeatable pattern, with the maximum channel used in the pattern being an upper bound.

Lemma 2

If $k_1 = 2$ and $k_2 = 1$, then an upper bound on the span is nine.

Proof. Suppose $k_1 = 2$ and $k_2 = 1$. If a transmitter has the channel x, then no transmitters within distance $2s$ can have the channels $x - 1$, x, or $x + 1$. By attempting to assign channels to the hexagons in a three-by-three grid with a lower maximum channel number, we find the pattern in Figure 2.

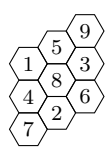


Figure 2: Channel assignment upper bound for a three-by-three grid with $k_1 = 2$ and $k_2 = 1$

We can see that the numbering satisfies the two constraints given by k_1 and k_2. This pattern can be repeated as depicted in Figure 3.

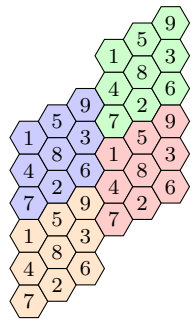


Figure 3: Repeated channel assignment from Figure 2

Note that the numbering satisfies the constraints across the boundaries of the shaded regions. This pattern can be repeated infinitely, so an upper bound for $k_1 = 2$ and $k_2 = 1$ is nine. ⬢

For the lower bound, we use a counting argument.

Lemma 3

If $k_1 = 2$ and $k_2 = 1$, then a lower bound on the span is nine.

Proof. Suppose $k_1 = 2$. Then, no adjacent hexagons can have a difference in values of less than two units. By Lemma 1, any optimal channel assignment must contain some hexagon assigned 1. Consider the pattern in Figure 4, in which there must be a transmitter assigned at least 8 because the assignments around 1 are distinct.

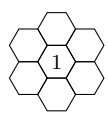


Figure 4: First step

Suppose, for contradiction, that the maximum channel assignment is 8. Then, in the pattern in Figure 5, 8 must be surrounded by 1, 2, 3, 4, 5, 6, or 7. Since we need six distinct assignments around 8, we assign either a 1 or a 2 around 8.

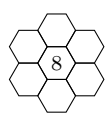


Figure 5: Second step

Without loss of generality, suppose we assign 1 or 2 to x. Consider those two cases with the pattern in Figure 6.

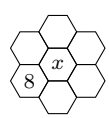


Figure 6: Assign x 1 or 2

Suppose $x = 1$. Then, we assign channels 3, 4, 5, or 6 to y in Figure 7.

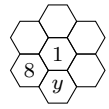


Figure 7: Assign y 3, 4, 5, or 6

We consider those assignments for y with the shaded hexagons in Figure 8. If $y = 3$, then the possible assignments for the shaded hexagons are 5, 6, and 7. If $y = 4$, then the possible assignments are 2, 6, and 7. If $y = 5$, then the possible assignments are 2, 3, and 7. If $y = 6$, then the possible assignments are 2, 3, and 4. In all these cases we have three values to assign to four hexagons, which is impossible because the assignments must be distinct.

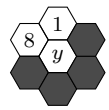


Figure 8: Hexagons to consider with cases for y

Suppose $x = 2$. Then, the possible assignments for the shaded hexagons in Figure 9 are 4, 5, 6, and 7, but there are five shaded hexagons. So, we need a higher channel number to assign, and the maximum channel assignment is not 8.

Figure 9: Hexagons to assign with $x = 2$

Then, there must be a channel assigned at least 9, so a lower bound when $k_2 = 1$ is nine.

Those lemmas result in the following theorem.

Theorem 1

If $k_1 = 2$ and $k_2 = 1$, then the span is nine.

Proof. Suppose $k_1 = 2$ and $k_2 = 1$. From Lemma 2 and Lemma 3, an upper bound and a lower bound on the span is nine.

5.2 $k_1 = 3$

We now determine upper and lower bounds on the span if $k_1 = 3$ using a repeatable pattern and a counting argument.

Lemma 4

If $k_1 = 3$ and $k_2 = 1$, then an upper bound on the span is 12.

Proof. Suppose $k_1 = 3$ and $k_2 = 1$. Then, if a transmitter has the channel x, then no transmitters within distance $2s$ can have the channels $x-2$, $x-1$, x, $x+1$, or $x+2$. Using a strategy similar to Lemma 2, we find the pattern in Figure 10.

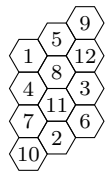


Figure 10: Channel assignment upper bound for a three-by-four grid with $k_1 = 3$ and $k_2 = 1$

This pattern satisfies the two constraints given by k_1 and k_2, and it can be repeated by extending horizontally and vertically without offset, like the $k_1 = 2$ case. The repeated pattern is in Figure 11, and the assignments satisfy the constraints across the shaded region boundary. So, an upper bound for $k_1 = 3$ and $k_2 = 1$ is 12.

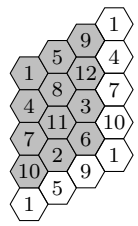


Figure 11: Repeated channel assignment from Figure 10

Lemma 5

If $k_1 = 3$ and $k_2 = 1$, then a lower bound on the span is 12.

Proof. Suppose $k_1 = 3$. Then, no adjacent hexagons can have a difference in values of less than three units. By Lemma 1, we can assume that a 1 exists somewhere in any optimal assignment. Thus, consider a pattern as shown in Figure 12, centered on a 1.

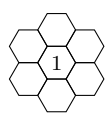


Figure 12: First step

The smallest number possible in any of the empty hexagons is 4. The number of each of the empty hexagons must be unique since any one of them is within two hexagons of all the others and $k_2 = 1$. Thus, the smallest possible span for this shape is 9.

Assume that our span is nine. Then, every number 4 through 9 must appear surrounding the 1, so a 5 must appear in our grid.

Since adjacent hexagons' numbers must be at least three apart, the only options to surround the 5 are 1, 2, 8, and 9. But there are six spaces surrounding the 5, so we must have at least six distinct options. Thus, our span must be at least 11. If our span is 11, we must have an 11 in our grid. The possible options to surround an 11 are $\{1, \ldots, 8\}$, so in any assignment, exactly six of those numbers must be selected to surround an 11. Thus, any 11 must have either a 3, 4, or 5 adjacent to it:

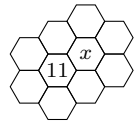


Figure 13: Second step ($x \in \{3, 4, 5\}$)

We consider the numbers possible to fill the five empty hexagons adjacent to x. If $x = 3$, then we have $6, 7, 8, 9, 10$, if $x = 4$, then we have $1, 7, 8, 9, 10$, and if $x = 5$, then we have $1, 2, 8, 9, 10$. In each case we have exactly five possible numbers, all of which must be used. None of 8, 9, or 10 can be placed adjacent to 11, so they must fill the other three hexagons, shaded in Figure 14:

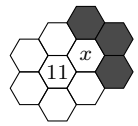


Figure 14: Third step ($x \in \{3, 4, 5\}$)

However, no two of the numbers 8, 9, and 10 can be placed adjacent to each other, thus they cannot fill these three hexagons. Therefore, we have reached a contradiction, so our span must be at least 12.

Theorem 2

If $k_1 = 3$ and $k_2 = 1$, then the span is 12.

Proof. Suppose $k_1 = 3$ and $k_2 = 1$. Combining Lemma 4 and Lemma 5, we have that an upper bound and a lower bound on the span is 12.

5.3 $k_1 \geq 4$

We again determine upper and lower bounds on the span for $k_1 \geq 4$. For the upper bound, we find a generalized repeatable pattern.

Lemma 6

If $k_1 \geq 4$ and $k_2 = 1$, then an upper bound on the span is $2k_1 + 7$.

Proof. Suppose $k_1 \geq 4$ and $k_2 = 1$. Then, if a transmitter has the channel x, then no transmitters within distance $2s$ can have any channel in $\{x - k_1 + 1, x - k_1 + 2, \ldots, x + k_1 - 2, x + k_1 - 1\}$. We find a repeatable pattern in a three-by-three grid. To find an upper bound, M, we assign channels to the hexagons as shown in Figure 15, with x, y, and z being unknown assignments.

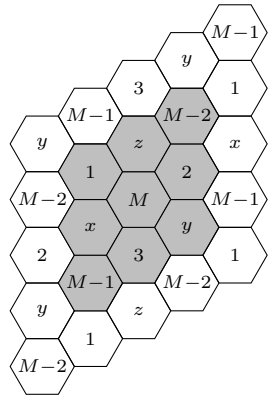


Figure 15: Repeated channel assignment for $k_1 \geq 4$ with x, y, and z.

For both constraints to hold, we require $M - 2 \geq 3 + k_1$ and $x, y, z \in \{k_1 + 3, k + 4, \ldots, M - 2 - k_1\}$. So, the smallest M satisfies $k_1 + 5 = M - 2 - k_1$, and $M = 2k_1 + 7$. ⬢

For the lower bound we consider more cases for k_1.

Lemma 7

If $k_1 \geq 4$ and $k_2 = 1$, then a lower bound on the span is $2k_1 + 7$.

Proof. Suppose $k_2 = 1$. We consider two cases: $4 \leq k_1 \leq 6$ and $k_1 \geq 7$.

Case 1: Suppose $4 \leq k_1 \leq 6$. Then no two adjacent hexagons may be assigned numbers within k_1 of each other. By Lemma 1, any optimal assignment must contain a 1. This 1 must be surrounded by six distinct numbers, each at least $1 + k_1$. Thus, the minimum span is at least $2k_1 + 4$, as in the arrangement shown in Figure 16:

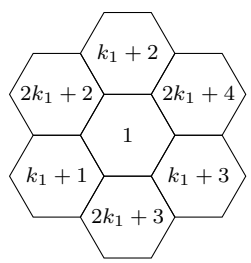


Figure 16: Minimized assignment

Note that while there are ways to rearrange the numbers in this arrangement, there is no arrangement that has a maximum below $2k_1 + 4$. There is also no way to arrange them that does not include a hexagon assigned $k_1 + 1$. Thus, an assignment with a span of $2k_1 + 4$ must include a $k_1 + 1$. However, the only $a \in \mathbb{Z}$ with $a \leq 2k_1 + 4$ and $|a - (k_1 + 1)| \geq k_1$ are $1, 2k_1 + 1, 2k_1 + 2, 2k_1 + 3, 2k_1 + 4$. Thus, it is impossible to assign all six hexagons around a $k_1 + 1$ with a span of $2k_1 + 4$, so our span must be at least $2k_1 + 5$.

Now, if our span is $2k_1 + 5$, there must be some hexagon assigned the value $2k_1 + 5$. Only the numbers 1 through $k_1 + 5$ can surround a $2k_1 + 5$, so one of the numbers in $\{3, 4, \ldots, k_1 + 2\}$ must be adjacent to any $2k_1 + 5$.

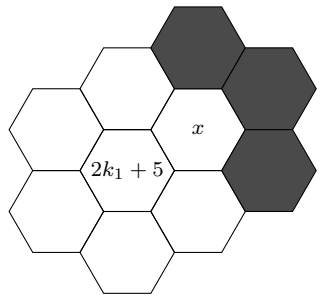


Figure 17: Assignment for span $= 2k_1 + 5$ with $x \in \{3, 4, \ldots k_1 + 2\}$

The possible options to surround any x are $\{1, \ldots, x - k_1\} \cup \{x + k_1, \ldots, 2k_1 + 4\}$. The maximum value of $x - k_1$ is 2, so there are at most two possible values less than k_1. This happens only if x is $k_1 + 1$ or $k_1 + 2$, and then the values in $\{x + k_1, \ldots, 2k_1 + 4\}$ are at most four sequential values, meaning that none of them can be adjacent to each other. Since all but at most one of them cannot be adjacent to the $2k_1 + 5$ in the middle, it is impossible to create a valid assignment. Thus, we consider smaller values of x. These are at most $2k_1 + 4 - (k_1 + 3) + 1 = k_1 + 2$ possible values above k_1, all of which are sequential. Values must be k_1 apart in order to be adjacent, so at most four of these values can possibly be adjacent to any other in the set. Thus, it is impossible to create a valid assignment with any of these values of x, and therefore our span must be at least $2k_1 + 6$.

So, any assignment must contain a hexagon with value $2k_1 + 6$. The possible values to surround it are $1, \ldots, k_1 + 6$, so at least one of the values in $\{4 \ldots, k_1 + 2\} \cup \{k_1 + 5, k_1 + 6\}$ must be adjacent to any $2k_1 + 6$, see Figure 18.

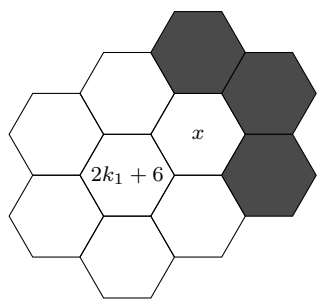


Figure 18: Assignment for span $= 2k_1 + 6$ with $x \in \{4 \ldots, k_1 + 2\} \cup \{k_1 + 5, k_1 + 6\}$

If $x \in \{4, \ldots, k_1 + 2\}$, then we consider the candidates $\{1, \ldots, x - k_1\} \cup \{x + k_1, \ldots, 2k_1 + 5\}$ to surround x. If $x \leq k_1$, then this set consists of only $\{x + k_1, \ldots, 2k_1 + 5\}$, a set of at most $k_1 + 2$ sequential numbers. Since numbers must be at least k_1 apart to be assigned to adjacent hexagons, this means that at most four of those numbers can possibly be assigned next to any other. Since there are five spaces surrounding the x, that means there can be no valid assignment. If $x \geq k_1 + 1$, then there are at most two possible values smaller than k_1, and at most five possible values greater than k_1. These are sequential, and thus no more than two of them can be adjacent to any other, and none can be adjacent to $2k_1 + 6$. Thus, it is impossible to arrange these values into a valid assignment.

If $x = k_1 + 5$, then the candidates to surround x consist of $2k_1 + 5$ and $\{1, \ldots, 5\}$. At most, 1 and 5 from the smaller set can be adjacent to one another, and any of them can be adjacent to $2k_1 + 5$. This does not allow us to fill five spaces without overlap, thus a valid assignment is impossible.

If $x = k_1 + 6$, then the candidates to surround x consist of $\{1, \ldots, 6\}$. At most four of these can be adjacent to any other, so it is impossible to create a valid assignment for the five empty spaces.

Since no value of x gives us a valid assignment with a span of $2k_1 + 6$, the span must be at least $2k_1 + 7$.

Case 2: Suppose $k_1 \geq 7$. Then no two adjacent hexagons may be assigned numbers within k_1 of each other. By

Lemma 1, any optimal assignment must contain a 1. The minimum value for any hexagon adjacent to this 1 is $k_1 + 1$, and the minimum for any hexagon adjacent to both of those is $2k_1 + 1$, as shown in Figure 19.

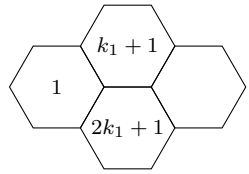


Figure 19: Second step *(all values shown are minimums, not assignments)*

If these hexagons are assigned to the values shown above, the smallest possible value for the empty hexagon is $3k_1 + 1$, because any smaller value is too close to one of the other three hexagons. However, we have proven that the upper bound for the minimum span is $2k_1 + 7$, and since $k_1 \geq 7$, we have $3k_1 + 1 > 2k_1 + 7$. Thus we increase our minimum by one, allowing us to assign 2 to the empty space, see Figure 20.

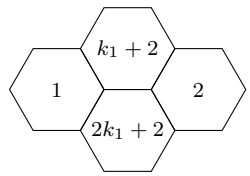


Figure 20: Third step

Consider the following shape.

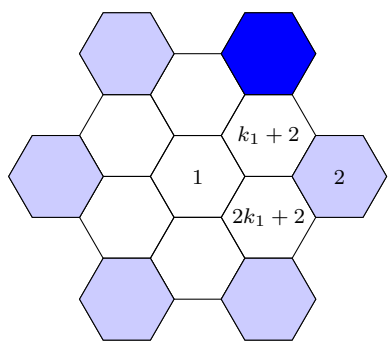


Figure 21: Fourth step

In this arrangement, every hexagon in the first ring around the 1 must be assigned a value at least $k_1 + 1$. From the previous steps, we see that assigning a value of $k_1 + 1$ produces a contradiction when assigning the second ring (blue in the above figure). Similarly, assigning $k_1 + 2$ produces the same contradiction if we attempt to assign the second adjacent element of the second ring (the dark blue hexagon in the above figure). Thus, each hexagon in the inner ring must be at least $k_1 + 3$. Because adjacent hexagons must be at least k_1 apart, values of the form $k_1 + a$ and $2k_1 + a$ must alternate around the inner ring. Thus, there must be three of the form $k_1 + a$, with distinct a at least 3. Thus, their minimum values are a equal to 3, 4, and 5, see Figure 22.

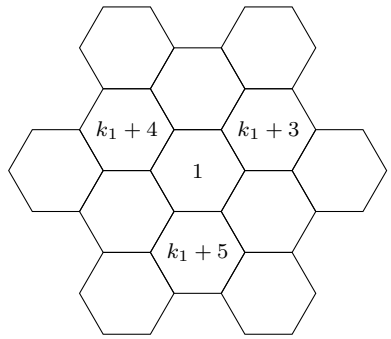


Figure 22: Fifth step

This results in minimum values for the other three hexagons in the inner circle of $2k_1 + a$ with a equal to 4, 5, and 6, see Figure 23.

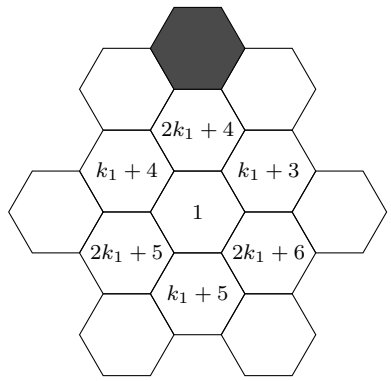


Figure 23: Sixth step

The shaded hexagon can only be filled with $k_1 + 1$ or $k_1 + 2$, either of which gives us too few options for the two empty hexagons on either side of it. Thus, each of the $2k_1 + a$ hexagons in the inner circle must have $a \geq 5$. So, we must reassign as shown in Figure 24.

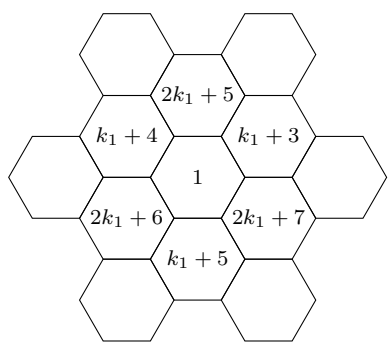


Figure 24: Seventh step

Thus, in order to create a valid assignment we must have a span of at least $2k_1 + 7$.

Now we can prove the following theorem.

Theorem 3

If $k \geq 4$ and $k_2 = 1$, then the span is $2k_1 + 7$.

Proof. Suppose $k \geq 4$ and $k_2 = 1$. Then, by Lemma 6 and Lemma 7, an upper bound and a lower bound on the span is $2k_1 + 7$. ⬢

6 Finite Grids

We now look at the spans of finite grids. Note that we cannot directly apply the results from the infinite grid using the second observation that if one arrangement, A_1, of hexagons contains another arrangement, A_2, of hexagons, then the span of A_1 is at least the span of A_2. For example, the infinite grid contains the grid in Figure 25, but the span of the grid in Figure 25 with $k_1 = 2$ and $k_2 = 1$ is one, not nine. We can, however, use the upper bounds found for the infinite grid and any assignments for grids we constructed for the lower bounds.

Figure 25: Grid contained in the infinite grid with span one

In particular, generalizing the $x = 2$ case in the proof of Lemma 3, the pattern in Figure 26 with $x \in \{2, 3, 4, 5, 6, 7\}$ has span nine.

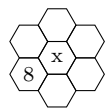


Figure 26: Generalized $x = 2$ case assignments

We also notice that without assigning 8, the pattern has span nine because there are six hexagons which we can assign channels, and there are $|\{1, \ldots, x-2, x+2, \ldots, 8\}| = 5$ possible channels. So any grid containing the assignment in Figure 27 has a span of at least nine, and applying the upper bound, any grid containing that assignment has span nine.

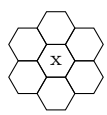


Figure 27: Assignment with span nine

7 Finite Grid with $k_1 = 2$ and $k_2 = 1$

We have the finite grid in Figure 28, and we want to determine the span with $k_1 = 2$ and $k_2 = 1$.

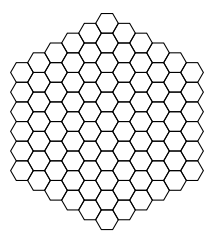


Figure 28: Finite grid to consider

Theorem 4

The span of the finite grid in Figure 28 with $k_1 = 2$ and $k_2 = 1$ is nine.

Proof. Suppose $k_1 = 2$ and $k_2 = 1$. Then, we know that the span is at most nine by Lemma 2. We determine whether there is a lower bound less than nine. It suffices to show that the grid in Figure 28 contains the grid and assignments in Figure 27. Assigning channels so that a $2, 3, 4, 5, 6$, or 7 does not end up on the inside of the grid, we have three possible channels for the hexagons not along the outside. For any assignments for the grid in Figure 26, all seven values must be distinct to satisfy both constraints, so we require more than three different channels inside the grid. The pattern in Figure 27 exists in Figure 28, so the span of the grid in Figure 28 is nine. ⬣

8 Infinite Grid with $k_1 = 2$ and $k_2 = 1$

We determine the span with $k_1 = 2$ and $k_2 = 1$ for the infinite grid. From Theorem 1, the span is nine.

9 Finite and Infinite Grids with $k_1 \geq 2$ and $k_2 = 1$

We find the span for the grid in Figure 28 and the infinite grid with generalized k_1.

9.1 Generalized Finite Grid

We first determine the span of the grid in Figure 28.

Theorem 5

If $k_1 \geq 2$ and $k_2 = 1$, then the span is nine if $k_1 = 2$, 12 if $k_1 = 3$, and $2k_1 + 7$ if $k_1 \geq 4$.

Proof. Suppose $k_1 \geq 2$ and $k_2 = 1$.

Case 1: Suppose $k_1 = 2$. By Theorem 4, the span is nine.

Case 2: Suppose $k_1 = 3$. Using the arrangement in Figure 14, the span of the grid in Figure 28 is 12 if there is an adjacent 11 and 3, 4, or 5 somewhere not on the outside of the grid in Figure 28. We expand these cases so that x in Figure 14 can also be 6, 7, 8, or 9. If $x = 6$, then the possible channels are $1, 2, 3, 9$, and 10. The 9 and 10 cannot be adjacent to 11, and for the remaining three hexagons adjacent to x, two of $1, 2$, or 3 are adjacent, which is impossible because $k_1 = 3$. If $x = 7$, then the possible channels are $1, 2, 3, 4$, and 10. Then, two pairs out of $1, 2, 3$, and 4 are adjacent, which is also impossible. If $x = 8$, then the possible channels are $1, 2, 3, 4$, and 5, and if $x = 9$, then the possible channels are $1, 2, 3, 4, 5$, and 6. In all cases, there are four adjacent pairs out of those channels, which is impossible.

For all the cases of x, there are exactly five possible channels, and there are six if we include 11. So, it is impossible to assign those six channels without introducing another channel. If the pattern in Figure 14 appears in Figure 28, not on the border, where $x \in \{3, 4, 5, 6, 7, 8, 9\}$, then the grid in Figure 28 has span at least 12. We attempt to assign channels so that $3, 4, 5, 6, 7, 8$, and 9 are only on the border. Then, $1, 2, 10$, and 11 can be inside the grid. But, as we noticed in the span for the grid in Figure 28, we need at least seven distinct values inside the grid, so at least one of $3, 4, 5, 6, 7, 8$, or 9 must be inside the grid. Then, a lower bound on the span is 12.

By Lemma 4, an upper bound on the span is 12, so the span is 12.

Case 3: Suppose $k_1 \geq 4$. The finite grid in Figure 28 contains the shape from Figure 24, so by the same logic as the proof of Lemma 7, in which that figure appears, the span must attain a value of $2k_1 + 7$. By Lemma 6, an upper bound on the span is $2k_1 + 7$, so the span is $2k_1 + 7$. ⬣

9.2 Generalized Infinite Grid

We now apply theorems for the span of an infinite grid. From Theorem 1, if $k_1 = 2$, then the span is nine. From Theorem 2, if $k_1 = 3$, then the span is 12. From Theorem 3, if $k \geq 4$, then the span is $2k_1 + 7$.

10 Finite and Infinite Grids with Generalized k_2

We generalize the constraint on transmitters within $4s$ of each other for infinite grids. The previous sections model $k_2 = 1$, so instead, we consider $k_1 \geq k_2 \geq 1$. We find the span for $k_1 = k_2$ and find lower and upper bounds for $k_2 < k_1$.

Theorem 6

If $k_1 = k_2$, then the span is $6k_2 + 1$.

Proof. Suppose $k_1 = k_2$. By Lemma 1, any optimal channel assignment contains some hexagon assigned 1. Consider the pattern in Figure 29.

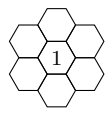


Figure 29: Assignment contains 1

To satisfy the constraint that channels for transmitters within distance $4s$ must differ by at least k_2, all the assignments in Figure 29 are distinct. Since channels for adjacent transmitters must differ by at least k_1, any minimized assignment uses the channels $1, k_1 + 1, \ldots, 6k_1 + 1$. Then, any minimized channel assignment for a grid containing the pattern in Figure 29 must have a channel assigned at least $6k_1+1$. So, if $k_1 = k_2$, then the span is at least $6k_2+1$.

Assign channels to hexagons as described in Figure 30.

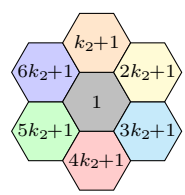


Figure 30: A minimized assignment pattern

Extending this pattern in Figure 31, we notice that no two hexagons with the same color are within $4s$ of each other. Since we can repeat this pattern, an upper bound for the span is $6k_2 + 1$.

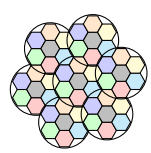


Figure 31: Extended minimized assignment pattern

So, if $k_1 = k_2$, the span is $6k_2 + 1$. ⬣

Theorem 7

If $k_1 \geq k_2$, then an upper bound on the span is $2k_1 + 6k_2 + 1$.

Proof. Similar to the proof of an upper bound with $k_1 \geq 4$ and $k_2 = 1$, we find a repeatable pattern in a three-by-three grid. To find an upper bound, M, we start by assigning the lowest and highest channels, which we depict

in Figure 32, like the previous proof of Lemma 7. Note that we need $M \geq 4k_2 + k_1 + 1$ so that adjacent channel assignments differ by at least k_1 as we repeat this pattern.

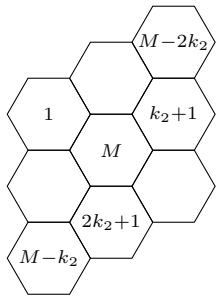


Figure 32: Assigning the lowest and highest values

Assigning variables to the unassigned transmitters and repeating the pattern, we get the assignment in Figure 33.

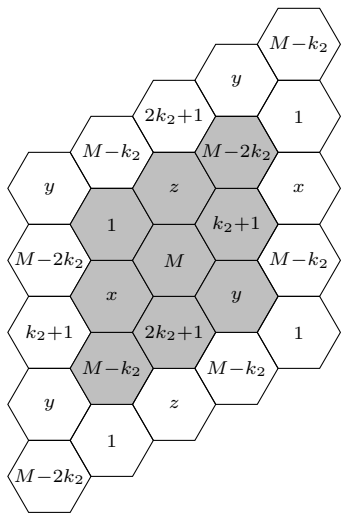


Figure 33: Repeated channel assignment for $k_1 \geq k_2$ with x, y, and z

Then, to satisfy the constraints, we need $x, y, z \in \{k_1 + 2k_2 + 1, k_1 + 2k_2 + 2, \ldots, M - 2k_2 - 1\}$. Also, x, y, and z must differ by at least k_2. Choosing the smallest assignments for x, y, and z, we use the channels $k_1 + 2k_2 + 1, k_1 + 3k_2 + 1$, and $k_1 + 4k_2 + 1$. We then solve for the smallest M so that $M - 2k_2 - 1 \geq k_1 + 4k_2 + 1$, resulting in $M \geq k_1 + 6k_2 + 2$. ⬣

Theorem 8

If $k_1 \geq 2k_2$, then a lower bound on the span is $2k_1 + 3k_2 + 1$, and if $2k_2 > k_1 \geq k_2$, then a lower bound on the span is $2k_1 + 4k_2 + 1$

Proof. Suppose $k_1 \geq k_2$. From Lemma 1 there is some hexagon assigned channel 1. Consider the pattern in Figure 34.

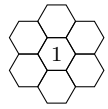


Figure 34: Assignment contains 1

The lowest value any of the unassigned hexagons could have is $k_1 + 1$, which we can assign to any hexagon. The next lowest value is $k_1 + k_2 + 1$ because the transmitters in all the hexagons are within $4s$ of each other, and it can be assigned to either of the shaded hexagons in Figure 35.

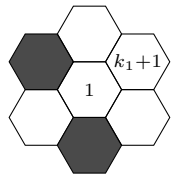


Figure 35: Assignment contains 1

Without loss of generality, assign it to the hexagon below the hexagon assigned 1. The next lowest value is $k_1 + 2k_2 + 1$, which we can only assign to the other shaded hexagon from Figure 35. We then have the assignments in Figure 36.

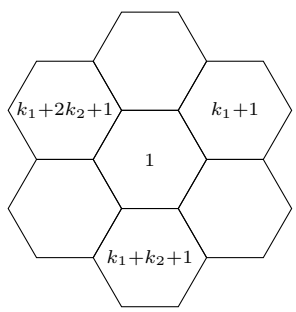


Figure 36: Assignment contains 1

Since the remaining unassigned hexagons are adjacent to hexagons with the coefficients 0 and 1 for k_1, the next lowest values are greater than $2k_1$. To ensure that they are at least k_2 away from any hexagon assigned, we find that the next lowest value is at least $2k_1 + k_2 + 1$.

If $k_1 \geq 2k_2$, then the next lowest value is $2k_1 + k_2 + 1$ and can only be assigned to the hexagon adjacent to both the $k_1 + 1$ and $k_1 + k_2 + 1$ hexagons. So, $2k_1 + 2k_2 + 1$ and $2k_1 + 3k_2 + 1$ can be assigned to the remaining two hexagons, resulting in a lower bound of $2k_1 + 3k_2 + 1$.

If $k_1 < 2k_2$, then the next lowest value is $2k_1 + 2k_2 + 1$ and can be assigned to any of the remaining hexagons. We then assign $2k_1 + 3k_2 + 1$ and $2k_1 + 4k_2 + 1$ to the other hexagons, resulting in a lower bound of $2k_1 + 4k_2 + 1$. ⬢

11 Conclusion

We prove a number of results for the minimum span. These include exact values for every case with $k_1 = 1$. We also prove a generalized upper bound for $k_1 \geq k_2$, a constraint that should always be true in the real world since transmitters close together should always have at least as much interference as those further apart. This upper bound is on the same order as all lower bounds we found, so it provides a reasonable bound on the number of channels that might be required to create an assignment. Possible future work on this problem could include working towards a better lower and upper bound, or even a definite formula, for the minimum span for the generalized case.